

Scaling ML-KEM to (much) higher security

Swechchha Adhikari¹, Vadim Lyubashevsky², and Peter Schwabe^{3,4}

¹ Brigham Young University, Provo, USA

mail@swechchhaa.com

² IBM Research, Zurich, Switzerland

VAD@zurich.ibm.com

³ Max Planck Institute for Security and Privacy, Bochum, Germany

peter@cryptojedi.org

⁴ Radboud University, Nijmegen, The Netherlands

1 Introduction

After a multi-year, multi-round public competition, NIST published its first post-quantum standards FIPS-203 [16], FIPS-204 [15], and FIPS-205 [17] in 2024, specifying ML-KEM, ML-DSA, and SLH-DSA respectively. Interestingly, four of the schemes in the PQC Round 3 Submissions [14] — Kyber [5] (standardized as ML-KEM), Saber [8], NTRU LPrime [3], and FrodoKEM [4] — are closely related lattice KEMs, spanning a spectrum of algebraic structure, from FrodoKEM’s unstructured LWE, through Kyber’s and Saber’s cyclotomic modules, to the prime-degree number fields of NTRU Prime. This spectrum reflects different design principles, commonly understood as structure vs conservatism. ML-KEM’s module structure gives small keys and fast NTT arithmetic, but is also seen as a potential attack surface, so conservative designs discard it entirely [4]. FrodoKEM, for example, is based on a more conservative hardness assumption of plain LWE, which is why bodies such as BSI, ENISA, and NLNCSA point to FrodoKEM for long-term confidentiality [18]; it was also recently standardized by ISO [11] for this very reason.

It is useful to separate the two notions of conservatism this framing bundles together as they concern two independent (and today, entirely hypothetical) classes of attack. The first arises from the choice of hardness assumption. FrodoKEM rests on unstructured LWE to protect against a structure break, an attack that would exploit the algebraic structure of ring/module LWE, but not apply to plain LWE. The second involves the security margin a scheme carries above its target level, which protects against advances in the best known attacks today. This could include improved lattice reduction, dual attacks, or sharper cost models that would affect the concrete security of every lattice scheme at its current parameters, including FrodoKEM. Both of these hedges are distinct and not to be substituted; structured conservatism does nothing against the second class of attacks and no security margin protects against the first. On top of that, it’s impossible to predict which of the two threats is more likely.

Faced with this uncertainty, increasing the margin has been a historically standard response. For computationally secure schemes, concrete security estimate is the cost of the best attack known at the time, and such estimates evolve

as cryptanalysis matures. For instance, the recommended size of an RSA modulus climbed from 512 to 1024 to 2048 bits and beyond as factoring algorithms and available computation advanced [2]. Lattice estimates are not static either. The cost models used for estimation and the attacks have evolved over more than a decade [7,1], and the cost of the best known attacks continues to drift in both directions as improvements are proposed [10,12], scrutinized [9], and refined [6].

If the best known attacks continue to improve, how much margin is required remains open, and the useful question is not whether today’s KEMs carry enough but how much margin they can carry at all. This question arises because additional margin comes at a cost, requiring larger parameters and hence higher bandwidth and runtime, so no scheme can indefinitely withstand advances in cryptanalysis simply by scaling its parameters. For LWE-family KEMs there is a further constraint, since the parameters that determine security also control the decryption failure probability, so increasing the noise to gain a larger security margin necessarily affects correctness.

Crucially, what structure determines is the cost of buying additional margin. At matched security levels, an unstructured scheme already requires roughly an order of magnitude more bandwidth than a module scheme [4,5]. Since these communication costs are considered practical enough for conservative deployments, a module scheme can spend the same bandwidth budget on larger parameters and obtain a conservative security margin while retaining the efficiency advantages of NTT arithmetic.

Recent work has explored this high-margin regime from the perspective of unstructured LWE. KyFrog [13] increases the LWE dimension and adjusts the noise distribution to achieve substantially higher security estimates for much larger ciphertexts. This motivates the complementary question of how much security margin a module KEM can achieve within the communication budget of a conservative unstructured scheme. We answer this question with KAIBURR, a module KEM that scales ML-KEM’s module rank and introduces a generalized noise distribution to preserve correctness. We show that KAIBURR achieves large security margins within FrodoKEM’s communication cost while running substantially faster.

1.1 Our contributions

1.2 Structure of the paper

1.3 Access to artifacts

All software, including parameter analysis scripts, implementation source code, and other artifacts, is publicly available on GitHub.

2 Preliminaries

3 Warmup: Just changing k

4 Revisiting the noise distribution

5 Benchmarks

Acknowledgements

Large parts of the work were done while the first author was employed at MPI-SP.

References

1. Albrecht, M.R., Player, R., Scott, S.: On the concrete hardness of learning with errors. *J. Math. Cryptol.* **9**(3), 169–203 (2015), <http://www.degruyter.com/view/j/jmc.2015.9.issue-3/jmc-2015-0016/jmc-2015-0016.xml> [2](#)
2. Barker, E.: Recommendation for key management: Part 1 — general. NIST Special Publication 800-57 Part 1, Revision 5, National Institute of Standards and Technology (May 2020). <https://doi.org/10.6028/NIST.SP.800-57pt1r5> [2](#)
3. Bernstein, D.J., Chuengsatiansup, C., Lange, T., van Vredendaal, C.: NTRU prime: Reducing attack surface at low cost. In: Adams, C., Camenisch, J. (eds.) *Selected Areas in Cryptography - SAC 2017 - 24th International Conference*, Ottawa, ON, Canada, August 16–18, 2017, Revised Selected Papers. *Lecture Notes in Computer Science*, vol. 10719, pp. 235–260. Springer (2017). https://doi.org/10.1007/978-3-319-72565-9_12 [1](#)
4. Bos, J.W., Costello, C., Ducas, L., Mironov, I., Naehrig, M., Nikolaenko, V., Raghunathan, A., Stebila, D.: Frodo: Take off the ring! practical, quantum-secure key exchange from LWE. In: *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. pp. 1006–1018. ACM (2016). <https://doi.org/10.1145/2976749.2978425> [1](#), [2](#)
5. Bos, J.W., Ducas, L., Kiltz, E., Lepoint, T., Lyubashevsky, V., Schanck, J.M., Schwabe, P., Seiler, G., Stehlé, D.: CRYSTALS - kyber: A CCA-secure module-lattice-based KEM. In: *2018 IEEE European Symposium on Security and Privacy, EuroS&P 2018*, London, United Kingdom, April 24–26, 2018. pp. 353–367. IEEE (2018). <https://doi.org/10.1109/EuroSP.2018.00032> [1](#), [2](#)
6. Carrier, K., Meyer-Hilfiger, C., Shen, Y., Tillich, J.: Assessing the impact of a variant of matzov’s dual attack on kyber. In: Kalai, Y.T., Kamara, S.F. (eds.) *Advances in Cryptology - CRYPTO 2025 - 45th Annual International Cryptology Conference*, Santa Barbara, CA, USA, August 17–21, 2025, *Proceedings, Part I*. *Lecture Notes in Computer Science*, vol. 16000, pp. 444–476. Springer (2025). https://doi.org/10.1007/978-3-032-01855-7_15, https://doi.org/10.1007/978-3-032-01855-7_15 [2](#)
7. Chen, Y., Nguyen, P.Q.: Bkz 2.0: Better lattice security estimates. In: Lee, D.H., Wang, X. (eds.) *Advances in Cryptology – ASIACRYPT 2011*. pp. 1–20. Springer Berlin Heidelberg, Berlin, Heidelberg (2011) [2](#)

8. D’Anvers, J.P., Karmakar, A., Roy, S.S., Vercauteren, F.: Saber: Module-LWR based key exchange, CPA-secure encryption and CCA-secure KEM. In: Joux, A., Nitaj, A., Rachidi, T. (eds.) AFRICACRYPT 18. LNCS, vol. 10831, pp. 282–305. Springer, Cham (May 2018). https://doi.org/10.1007/978-3-319-89339-6_16 1
9. Lucas, L., Pulles, L.N.: Does the dual-sieve attack on learning with errors even work? In: Handschuh, H., Lysyanskaya, A. (eds.) Advances in Cryptology - CRYPTO 2023 - 43rd Annual International Cryptology Conference, CRYPTO 2023, Santa Barbara, CA, USA, August 20–24, 2023, Proceedings, Part III. Lecture Notes in Computer Science, vol. 14083, pp. 37–69. Springer (2023). https://doi.org/10.1007/978-3-031-38548-3_2, https://doi.org/10.1007/978-3-031-38548-3_2 2
10. Guo, Q., Johansson, T.: Faster dual lattice attacks for solving LWE with applications to CRYSTALS. In: Tibouchi, M., Wang, H. (eds.) Advances in Cryptology - ASIACRYPT 2021 - 27th International Conference on the Theory and Application of Cryptology and Information Security, Singapore, December 6–10, 2021, Proceedings, Part IV. Lecture Notes in Computer Science, vol. 13093, pp. 33–62. Springer (2021). https://doi.org/10.1007/978-3-030-92068-5_2, https://doi.org/10.1007/978-3-030-92068-5_2 2
11. ISO/IEC: Information technology — security techniques — encryption algorithms — part 2: Asymmetric ciphers — amendment 2. Standard ISO/IEC 18033-2:2006/Amd 2:2026, International Organization for Standardization, Geneva, Switzerland (Jun 2026), <https://www.iso.org/standard/86890.html> 1
12. MATZOV: Report on the security of LWE: Improved dual lattice attack. Report, Zenodo (Apr 2022). <https://doi.org/10.5281/zenodo.6493704> 2
13. Melo, V.D., Buchanan, W.J.: Kyfrog: A high-security lwe-based kem inspired by ml-kem (2025), <https://arxiv.org/abs/2512.06411> 2
14. National Institute of Standards and Technology: Post-quantum cryptography: Round 3 submissions. <https://csrc.nist.gov/projects/post-quantum-cryptography/post-quantum-cryptography-standardization/round-3-submissions> (2020), accessed: 2026-07-27 1
15. National Institute of Standards and Technology: Module-lattice-based digital signature standard. Tech. Rep. Federal Information Processing Standards Publication (FIPS) 204, U.S. Department of Commerce, Washington, D.C. (Aug 2024). <https://doi.org/10.6028/NIST.FIPS.204>, <https://doi.org/10.6028/NIST.FIPS.204> 1
16. National Institute of Standards and Technology: Module-lattice-based key-encapsulation mechanism standard. Tech. Rep. Federal Information Processing Standards Publication (FIPS) 203, U.S. Department of Commerce, Washington, D.C. (Aug 2024). <https://doi.org/10.6028/NIST.FIPS.203>, <https://doi.org/10.6028/NIST.FIPS.203> 1
17. National Institute of Standards and Technology: Stateless hash-based digital signature standard. Tech. Rep. Federal Information Processing Standards Publication (FIPS) 205, U.S. Department of Commerce, Washington, D.C. (Aug 2024). <https://doi.org/10.6028/NIST.FIPS.205>, <https://doi.org/10.6028/NIST.FIPS.205> 1
18. The FrodoKEM Team: FrodoKEM: Practical quantum-secure key encapsulation from generic lattices. <https://frodokem.org/#news> (2026), news section. Accessed: 2026-07-27 1